

Power BI Assignment 2

1. Calculated Columns

A. Create a Calculated Column for 'Category Type': Add a calculated column in the Order Details table that combines the 'Category' and 'Sub-Category' columns into a single 'Category Type' column.

The screenshot shows the Power BI Desktop interface. The 'Column tools' ribbon is active, displaying options for Name, Data type, Format, Summarization, Data category, Sort, Data groups, Manage relationships, and New column. The 'Category Type' calculated column is being created with the formula: `1 Category Type = 2 'Order Details'[Category] & " - " & 'Order Details'[Sub-Category]`. The 'Data' pane on the right shows the 'Category Type' column added to the 'Order Details' table.

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type
B-25603	₹ 12	₹ 1	2	Clothing	Hankercchief	0.0833333333333333	Profit	Clothing - Hankercchief
B-25608	₹ 257	₹ 23	5	Clothing	Hankercchief	0.0894941634241245	Profit	Clothing - Hankercchief
B-25615	₹ 68	₹ 20	5	Clothing	Hankercchief	0.294117647058824	Profit	Clothing - Hankercchief
B-25616	₹ 42	₹ 12	5	Clothing	Hankercchief	0.285714285714286	Profit	Clothing - Hankercchief
B-25624	₹ 26	₹ 12	3	Clothing	Hankercchief	0.461538461538462	Profit	Clothing - Hankercchief
B-25625	₹ 97	₹ 29	2	Clothing	Hankercchief	0.298969072164948	Profit	Clothing - Hankercchief
B-25635	₹ 40	₹ 16	3	Clothing	Hankercchief	0.4	Profit	Clothing - Hankercchief
B-25638	₹ 154	₹ 39	3	Clothing	Hankercchief	0.253246753246753	Profit	Clothing - Hankercchief
B-25654	₹ 34	₹ 12	3	Clothing	Hankercchief	0.352941176470588	Profit	Clothing - Hankercchief
B-25656	₹ 6	₹ 3	1	Clothing	Hankercchief	0.5	Profit	Clothing - Hankercchief
B-25656	₹ 56	₹ 18	2	Clothing	Hankercchief	0.321428571428571	Profit	Clothing - Hankercchief
B-25670	₹ 24	₹ 1	2	Clothing	Hankercchief	0.0416666666666667	Profit	Clothing - Hankercchief
B-25670	₹ 14	₹ 2	1	Clothing	Hankercchief	0.142857142857143	Profit	Clothing - Hankercchief
B-25730	₹ 18	₹ 8	2	Clothing	Hankercchief	0.444444444444444	Profit	Clothing - Hankercchief
B-25751	₹ 32	₹ 7	3	Clothing	Hankercchief	0.21875	Profit	Clothing - Hankercchief
B-25757	₹ 106	₹ 15	7	Clothing	Hankercchief	0.141509433962264	Profit	Clothing - Hankercchief
B-25757	₹ 14	₹ 5	1	Clothing	Hankercchief	0.357142857142857	Profit	Clothing - Hankercchief
B-25757	₹ 17	₹ 7	3	Clothing	Hankercchief	0.411764705882353	Profit	Clothing - Hankercchief
B-25771	₹ 148	₹ 59	3	Clothing	Hankercchief	0.398648648648649	Profit	Clothing - Hankercchief
B-25783	₹ 26	₹ 2	2	Clothing	Hankercchief	0.0769230769230769	Profit	Clothing - Hankercchief
B-25784	₹ 15	₹ 4	1	Clothing	Hankercchief	0.266666666666667	Profit	Clothing - Hankercchief
B-25793	₹ 18	₹ 1	3	Clothing	Hankercchief	0.0555555555555556	Profit	Clothing - Hankercchief
B-25797	₹ 31	₹ 1	2	Clothing	Hankercchief	0.032258064516129	Profit	Clothing - Hankercchief
B-25800	₹ 45	₹ 12	7	Clothing	Hankercchief	0.266666666666667	Profit	Clothing - Hankercchief
B-25808	₹ 210	₹ 50	4	Clothing	Hankercchief	0.238095238095238	Profit	Clothing - Hankercchief
B-25809	₹ 154	₹ 54	3	Clothing	Hankercchief	0.350649350649351	Profit	Clothing - Hankercchief
B-25809	₹ 53	₹ 24	1	Clothing	Hankercchief	0.452830188679245	Profit	Clothing - Hankercchief

B. Calculate Revenue per Order in Order Details Table: Create a calculated column in the Order Details table to compute the revenue (Amount * Quantity) per order.

The screenshot shows the Power BI Desktop interface. The 'Column tools' ribbon is active, displaying options for Name, Data type, Format, Summarization, Data category, Sort, Data groups, Manage relationships, and New column. The 'Revenue per Order' calculated column is being created with the formula: `1 Revenue per Order = 2 'Order Details'[Amount] * 'Order Details'[Quantity]`. The 'Data' pane on the right shows the 'Revenue per Order' column added to the 'Order Details' table.

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type	Revenue per Order
B-25603	₹ 12	₹ 1	2	Clothing	Hankercchief	0.0833333333333333	Profit	Clothing - Hankercchief	₹ 24
B-25608	₹ 257	₹ 23	5	Clothing	Hankercchief	0.0894941634241245	Profit	Clothing - Hankercchief	₹ 1,285
B-25615	₹ 68	₹ 20	5	Clothing	Hankercchief	0.294117647058824	Profit	Clothing - Hankercchief	₹ 340
B-25616	₹ 42	₹ 12	5	Clothing	Hankercchief	0.285714285714286	Profit	Clothing - Hankercchief	₹ 210
B-25624	₹ 26	₹ 12	3	Clothing	Hankercchief	0.461538461538462	Profit	Clothing - Hankercchief	₹ 78
B-25625	₹ 97	₹ 29	2	Clothing	Hankercchief	0.298969072164948	Profit	Clothing - Hankercchief	₹ 194
B-25635	₹ 40	₹ 16	3	Clothing	Hankercchief	0.4	Profit	Clothing - Hankercchief	₹ 120
B-25638	₹ 154	₹ 39	3	Clothing	Hankercchief	0.253246753246753	Profit	Clothing - Hankercchief	₹ 462
B-25654	₹ 34	₹ 12	3	Clothing	Hankercchief	0.352941176470588	Profit	Clothing - Hankercchief	₹ 102
B-25656	₹ 6	₹ 3	1	Clothing	Hankercchief	0.5	Profit	Clothing - Hankercchief	₹ 6
B-25656	₹ 56	₹ 18	2	Clothing	Hankercchief	0.321428571428571	Profit	Clothing - Hankercchief	₹ 112
B-25670	₹ 24	₹ 1	2	Clothing	Hankercchief	0.0416666666666667	Profit	Clothing - Hankercchief	₹ 48
B-25670	₹ 14	₹ 2	1	Clothing	Hankercchief	0.142857142857143	Profit	Clothing - Hankercchief	₹ 14
B-25730	₹ 18	₹ 8	2	Clothing	Hankercchief	0.444444444444444	Profit	Clothing - Hankercchief	₹ 36
B-25751	₹ 32	₹ 7	3	Clothing	Hankercchief	0.21875	Profit	Clothing - Hankercchief	₹ 96
B-25757	₹ 106	₹ 15	7	Clothing	Hankercchief	0.141509433962264	Profit	Clothing - Hankercchief	₹ 742
B-25757	₹ 14	₹ 5	1	Clothing	Hankercchief	0.357142857142857	Profit	Clothing - Hankercchief	₹ 14
B-25757	₹ 17	₹ 7	3	Clothing	Hankercchief	0.411764705882353	Profit	Clothing - Hankercchief	₹ 51
B-25771	₹ 148	₹ 59	3	Clothing	Hankercchief	0.398648648648649	Profit	Clothing - Hankercchief	₹ 444
B-25783	₹ 26	₹ 2	2	Clothing	Hankercchief	0.0769230769230769	Profit	Clothing - Hankercchief	₹ 52
B-25794	₹ 15	₹ 4	1	Clothing	Hankercchief	0.266666666666667	Profit	Clothing - Hankercchief	₹ 15
B-25793	₹ 18	₹ 1	3	Clothing	Hankercchief	0.0555555555555556	Profit	Clothing - Hankercchief	₹ 54
B-25797	₹ 31	₹ 1	2	Clothing	Hankercchief	0.032258064516129	Profit	Clothing - Hankercchief	₹ 62
B-25800	₹ 45	₹ 12	7	Clothing	Hankercchief	0.266666666666667	Profit	Clothing - Hankercchief	₹ 315
B-25808	₹ 210	₹ 50	4	Clothing	Hankercchief	0.238095238095238	Profit	Clothing - Hankercchief	₹ 840
B-25809	₹ 154	₹ 54	3	Clothing	Hankercchief	0.350649350649351	Profit	Clothing - Hankercchief	₹ 462
B-25809	₹ 53	₹ 24	1	Clothing	Hankercchief	0.452830188679245	Profit	Clothing - Hankercchief	₹ 53

C. Create a Calculated Column to Categorize Sales: Add a calculated column named 'Sales Category' in the Order Details table that categorizes each order as 'Above Average' or 'Below Average' based on the Amount value.

The screenshot shows the Power BI Desktop interface. The 'Table tools' ribbon is active, and the 'Column tools' group shows the 'Sales Category' column has been added to the 'Order Details' table. The calculated column formula is visible in the formula bar: `IF('Order Details'[Amount] > AVERAGE('Order Details'[Amount]), 'Above Average', 'Below Average')`. The table view shows the 'Sales Category' column populated with 'Above Average' and 'Below Average' values for each row.

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type	Revenue per Order	Sales Category
B-25603	₹ 12	₹ 1	2	Clothing	Hankchief	0.0833333333333333	Profit	Clothing - Hankchief	₹ 24	Below Average
B-25608	₹ 257	₹ 23	5	Clothing	Hankchief	0.0894941634241245	Profit	Clothing - Hankchief	₹ 1,285	Below Average
B-25615	₹ 68	₹ 20	5	Clothing	Hankchief	0.294117647058824	Profit	Clothing - Hankchief	₹ 340	Below Average
B-25616	₹ 42	₹ 12	5	Clothing	Hankchief	0.285714285714286	Profit	Clothing - Hankchief	₹ 210	Below Average
B-25624	₹ 26	₹ 12	3	Clothing	Hankchief	0.461538461538462	Profit	Clothing - Hankchief	₹ 78	Below Average
B-25625	₹ 97	₹ 29	2	Clothing	Hankchief	0.298969072164948	Profit	Clothing - Hankchief	₹ 194	Below Average
B-25635	₹ 40	₹ 16	3	Clothing	Hankchief	0.4	Profit	Clothing - Hankchief	₹ 120	Below Average
B-25638	₹ 154	₹ 39	3	Clothing	Hankchief	0.253246753246753	Profit	Clothing - Hankchief	₹ 462	Below Average
B-25654	₹ 34	₹ 12	3	Clothing	Hankchief	0.352941176470588	Profit	Clothing - Hankchief	₹ 102	Below Average
B-25656	₹ 6	₹ 3	1	Clothing	Hankchief	0.5	Profit	Clothing - Hankchief	₹ 6	Below Average
B-25656	₹ 56	₹ 18	2	Clothing	Hankchief	0.321428571428571	Profit	Clothing - Hankchief	₹ 112	Below Average
B-25670	₹ 24	₹ 1	2	Clothing	Hankchief	0.0416666666666667	Profit	Clothing - Hankchief	₹ 48	Below Average
B-25670	₹ 14	₹ 2	1	Clothing	Hankchief	0.142857142857143	Profit	Clothing - Hankchief	₹ 14	Below Average
B-25730	₹ 18	₹ 8	2	Clothing	Hankchief	0.444444444444444	Profit	Clothing - Hankchief	₹ 36	Below Average
B-25751	₹ 32	₹ 7	3	Clothing	Hankchief	0.21875	Profit	Clothing - Hankchief	₹ 96	Below Average
B-25757	₹ 106	₹ 15	7	Clothing	Hankchief	0.141509433962264	Profit	Clothing - Hankchief	₹ 742	Below Average
B-25757	₹ 14	₹ 5	1	Clothing	Hankchief	0.357142857142857	Profit	Clothing - Hankchief	₹ 14	Below Average
B-25757	₹ 17	₹ 7	3	Clothing	Hankchief	0.411764705882353	Profit	Clothing - Hankchief	₹ 51	Below Average
B-25771	₹ 148	₹ 59	3	Clothing	Hankchief	0.398648648648649	Profit	Clothing - Hankchief	₹ 444	Below Average
B-25783	₹ 26	₹ 2	2	Clothing	Hankchief	0.0769230769230769	Profit	Clothing - Hankchief	₹ 52	Below Average
B-25784	₹ 15	₹ 4	1	Clothing	Hankchief	0.266666666666667	Profit	Clothing - Hankchief	₹ 15	Below Average
B-25793	₹ 18	₹ 1	3	Clothing	Hankchief	0.0535555555555556	Profit	Clothing - Hankchief	₹ 54	Below Average

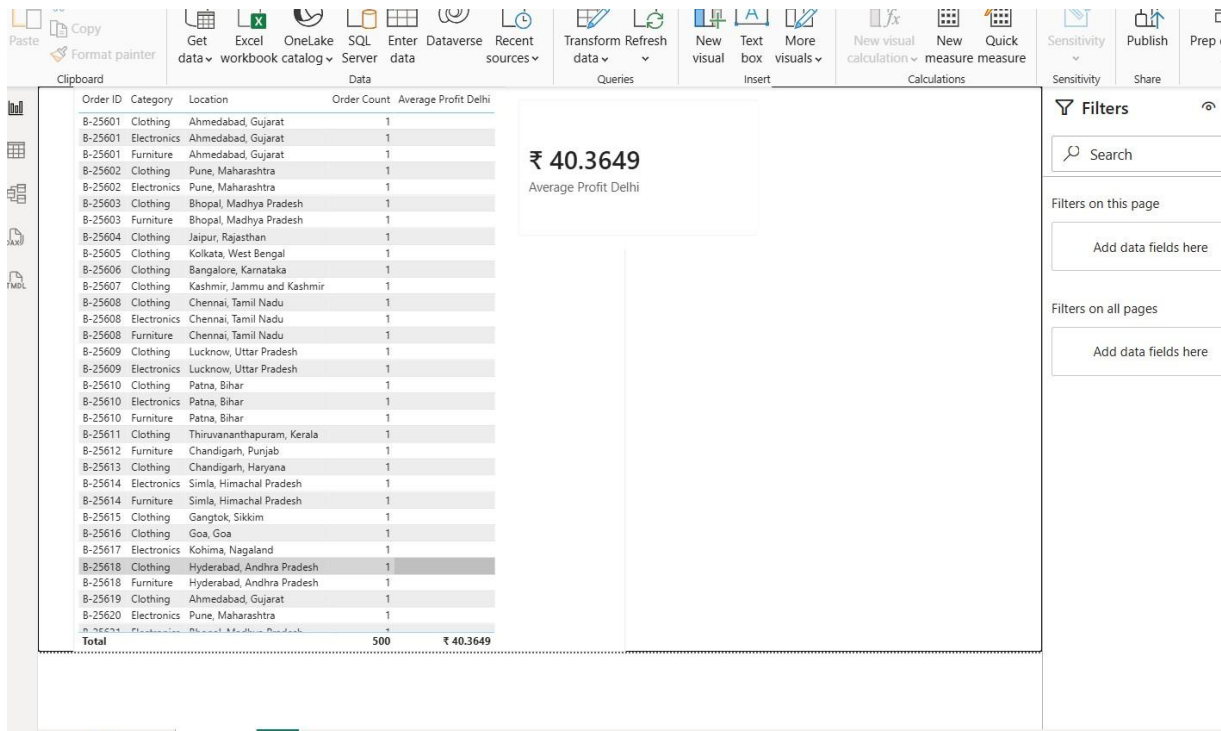
2. Calculated Measures

A. Order Count- Create a measure that calculates the total number of orders in the Order Details table.

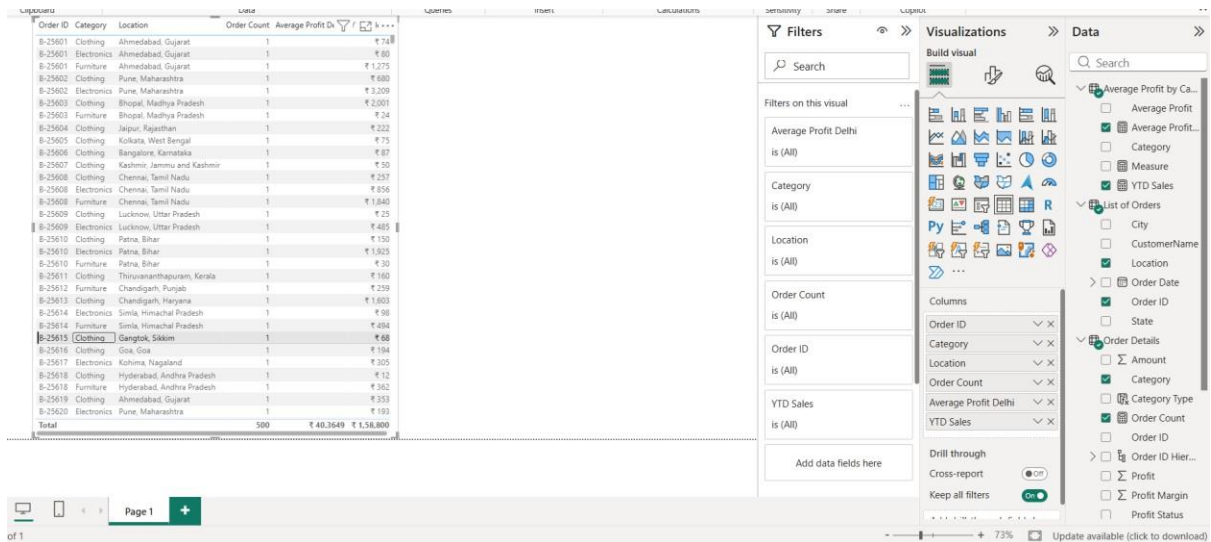
The screenshot shows the Power BI Desktop interface. The 'Visualizations' pane on the right shows a new measure 'Order Count' has been added to the 'Order Details' table. The measure is calculated as `COUNT([Order ID])`. The table view shows the 'Order Count' column populated with the total number of orders for each row.

Order ID	Category	Location	Order Count
B-25601	Clothing	Ahmedabad, Gujarat	1
B-25601	Electronics	Ahmedabad, Gujarat	1
B-25601	Furniture	Ahmedabad, Gujarat	1
B-25602	Clothing	Pune, Maharashtra	1
B-25602	Electronics	Pune, Maharashtra	1
B-25603	Clothing	Bhopal, Madhya Pradesh	1
B-25603	Furniture	Bhopal, Madhya Pradesh	1
B-25604	Clothing	Jaipur, Rajasthan	1
B-25605	Clothing	Kolkata, West Bengal	1
B-25606	Clothing	Bangalore, Karnataka	1
B-25607	Clothing	Kashmir, Jammu and Kashmir	1
B-25608	Clothing	Chennai, Tamil Nadu	1
B-25608	Electronics	Chennai, Tamil Nadu	1
B-25608	Furniture	Chennai, Tamil Nadu	1
B-25609	Clothing	Lucknow, Uttar Pradesh	1
B-25609	Electronics	Lucknow, Uttar Pradesh	1
B-25610	Clothing	Patna, Bihar	1
B-25610	Electronics	Patna, Bihar	1
B-25610	Furniture	Patna, Bihar	1
B-25611	Clothing	Thiruvananthapuram, Kerala	1
B-25612	Furniture	Chandigarh, Punjab	1
B-25613	Clothing	Chandigarh, Haryana	1
B-25614	Electronics	Simla, Himachal Pradesh	1
B-25614	Furniture	Simla, Himachal Pradesh	1
B-25615	Clothing	Gangtok, Sikkim	1
B-25616	Clothing	Goa, Goa	1
B-25617	Electronics	Kohima, Nagaland	1
B-25618	Clothing	Hyderabad, Andhra Pradesh	1
B-25618	Furniture	Hyderabad, Andhra Pradesh	1
B-25619	Clothing	Ahmedabad, Gujarat	1
B-25620	Electronics	Pune, Maharashtra	1
Total			500

B. Average Profit in Delhi- Create a measure that calculates the average profit for orders placed in Delhi.



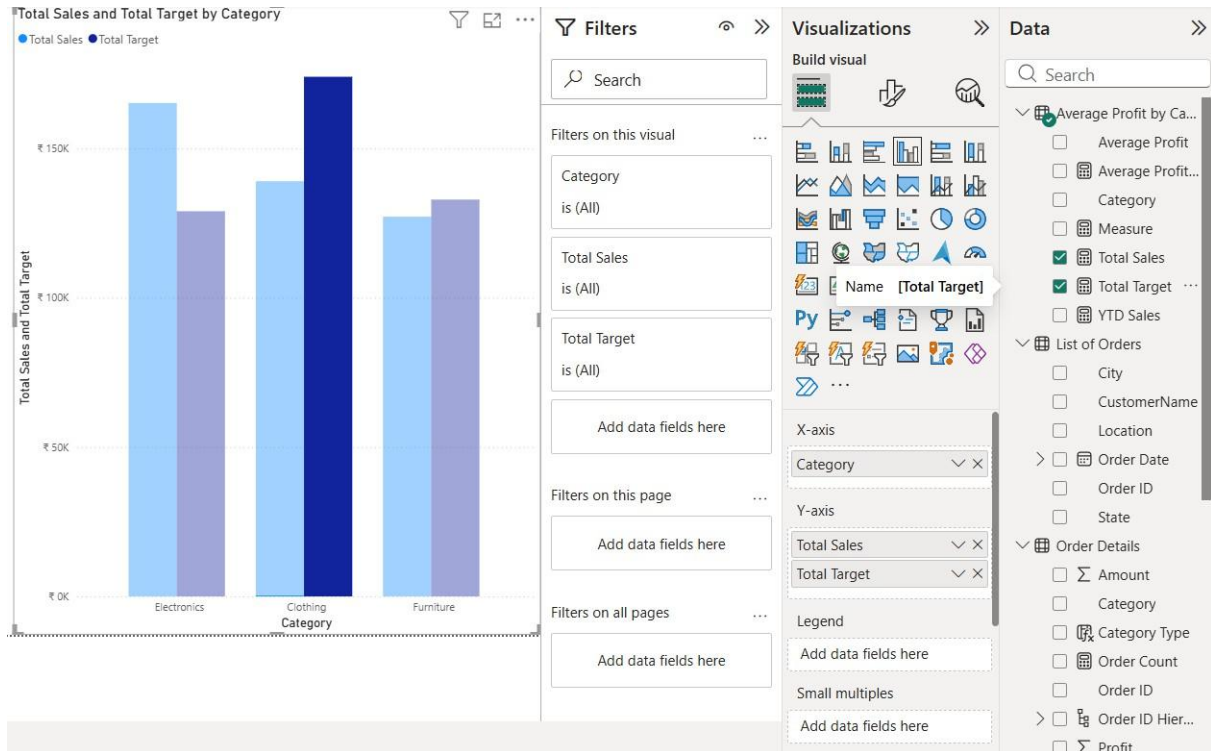
C. Year-to-Date (YTD) Sales- Create a measure that calculates total sales accumulated from the beginning of the year up to the current order date.



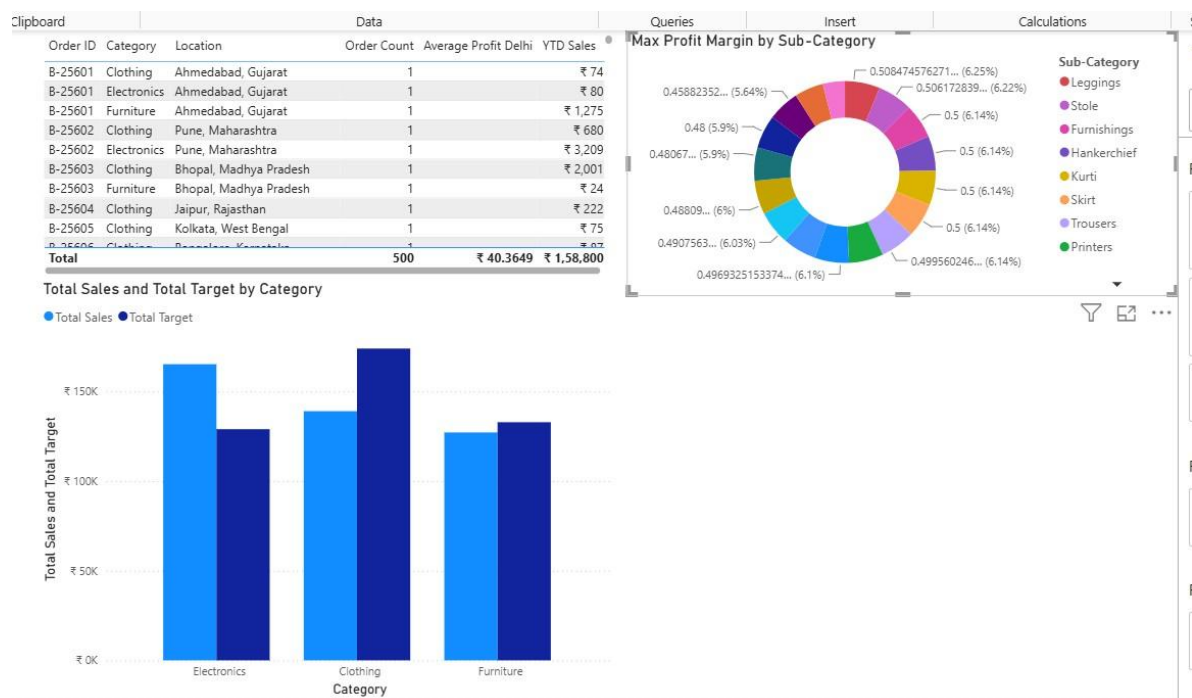
3. Data Visualization

Create the following visualizations using Power BI:

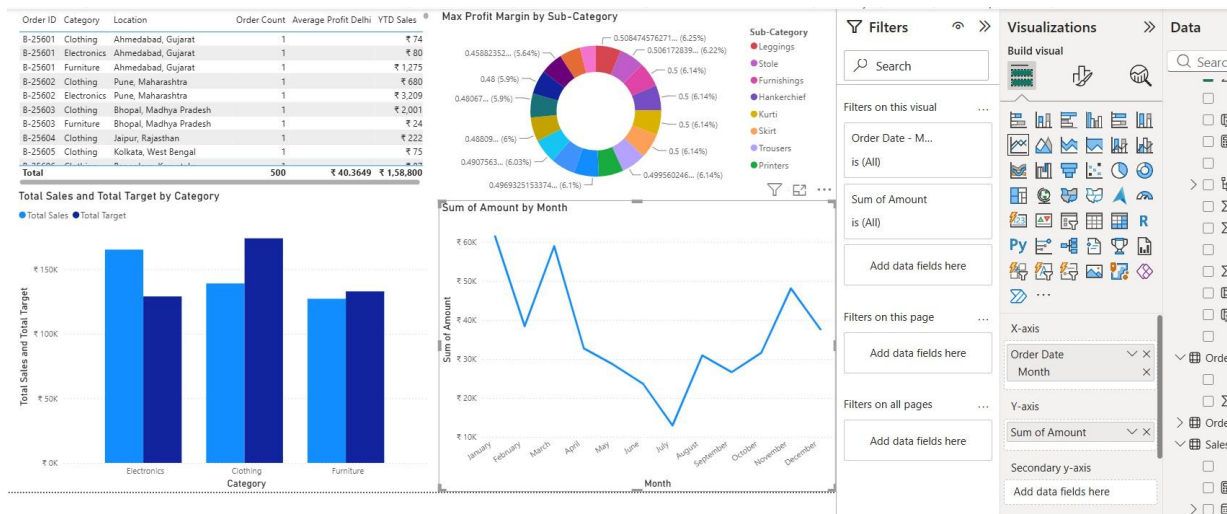
- A. Sales Target Achievement by Category: Compare actual sales with sales targets by category using a clustered column chart.



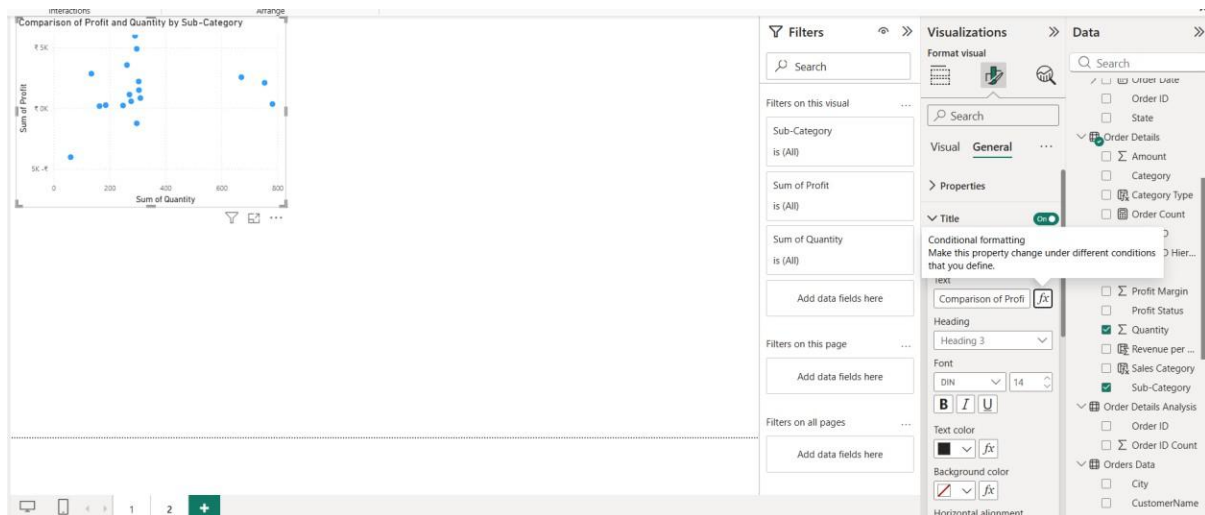
- B. Max Profit Margin by Sub-Category: Analyze the maximum profit margin for each sub-category of products using a donut chart.



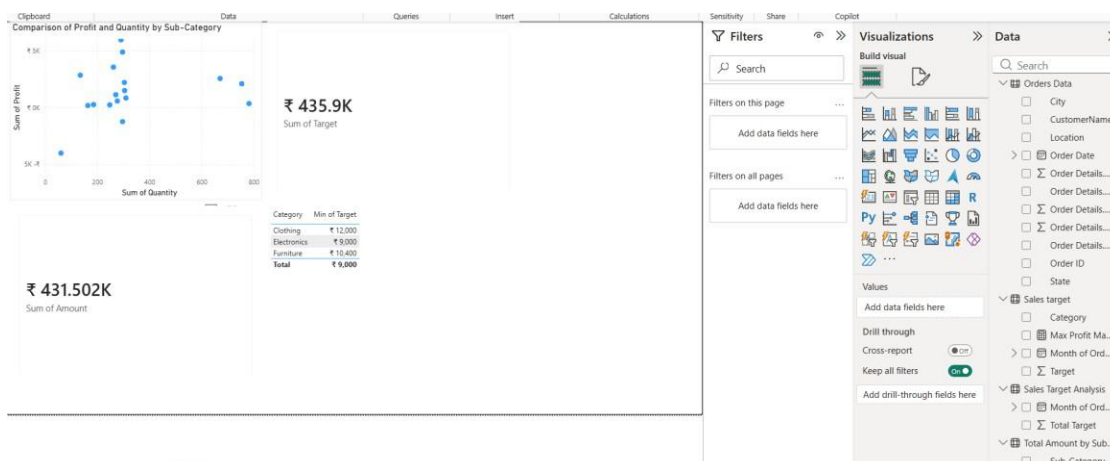
C. Monthly Sales Trend: Show the trend of monthly sales over time using a line chart.



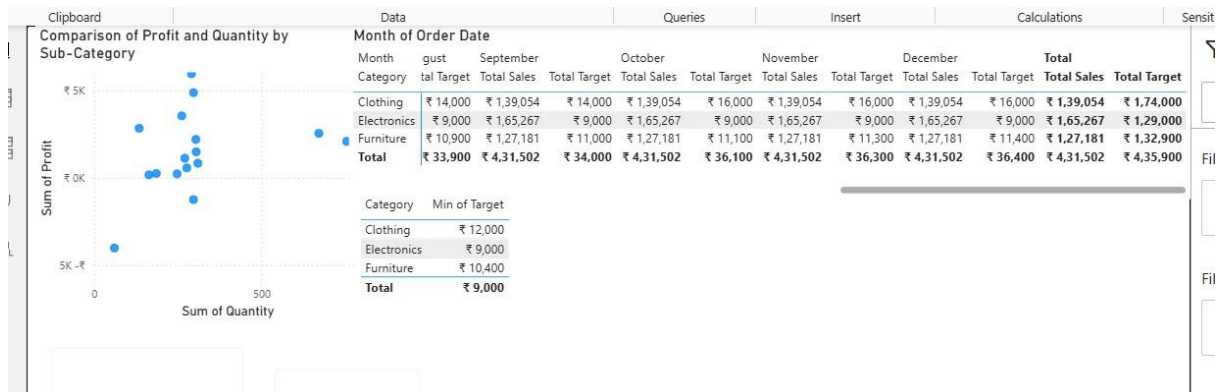
D. Comparison of Profit and Quantity by Sub-Category: Compare the relationship between profit and quantity sold for different sub-categories using a scatter chart.



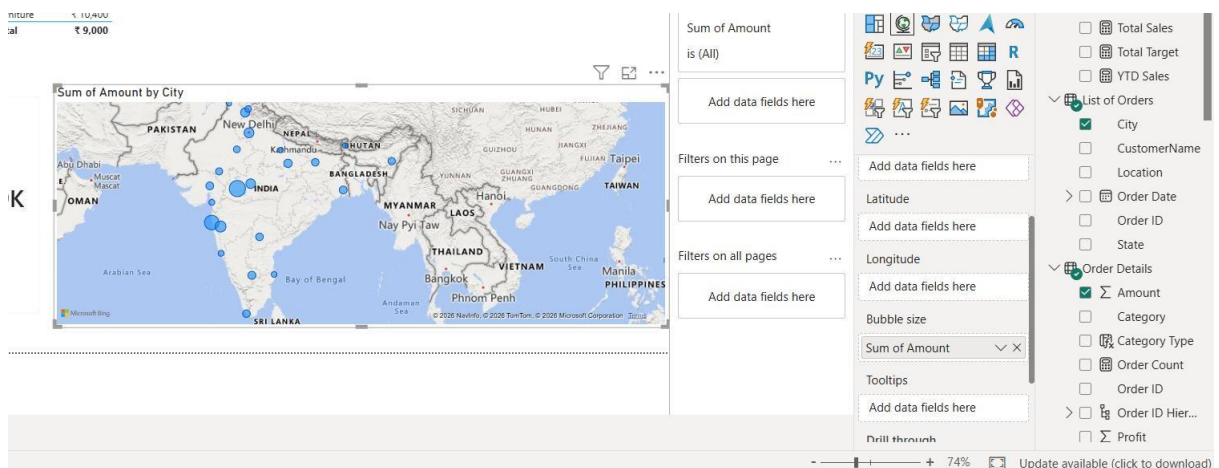
E. Comparison of Total Sales Amount and Target: Create cards to succinctly display the total sales amount alongside the sales target for quick comparison and analysis. Also, create a multi-row card to display the minimum target for each segment.



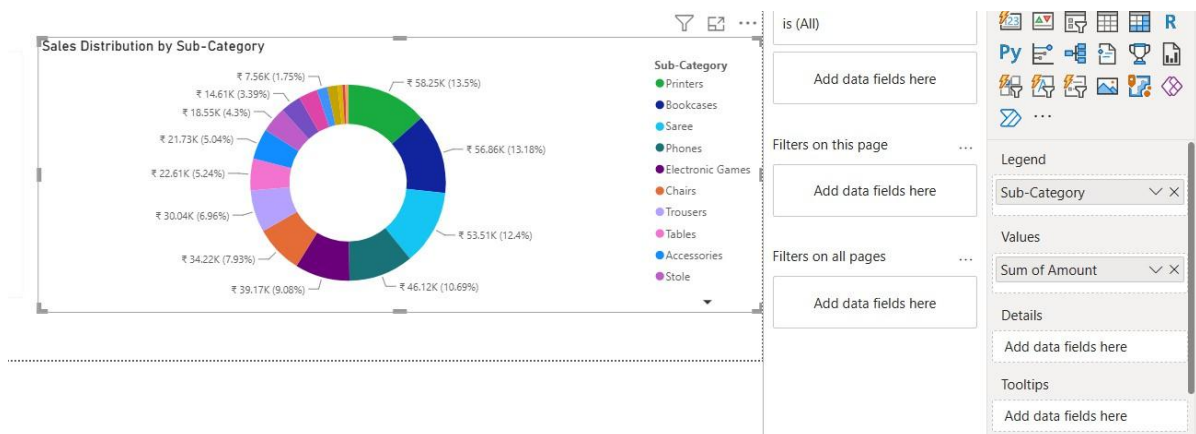
F. Sales Performance Matrix: Build a matrix view to analyze how actual sales compare to sales targets across different categories and months.



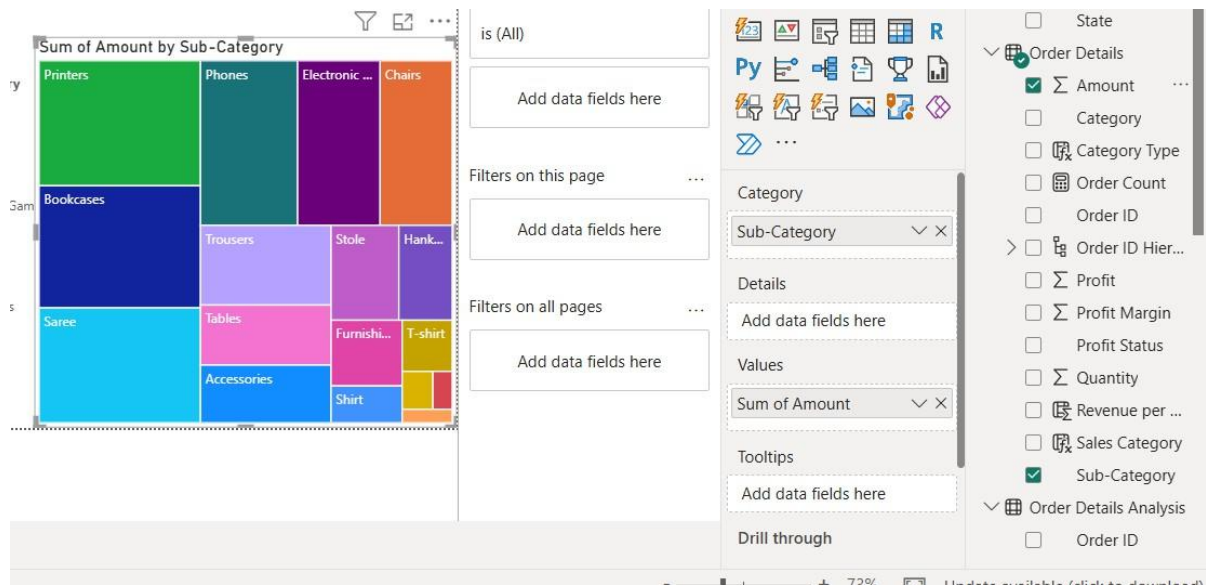
G. Geographic Sales Analysis: Visualize total sales on a map by city to identify regional sales patterns.



H. Sales Distribution by Sub-Category: Represent the sales distribution across different



I. Sub-categories using a tree map.



J. Order Count Analysis by State: Create a funnel chart to visualize the distribution of order counts across different states

